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Detailed Course

Introduction and Motivation

General Context

LDA (Linear Discriminant Analysis) is a **supervised latent analysis** method that finds the optimal latent factors for discrimination between classes.

Objectives of LDA:

- Introduce a **supervised** definition of latent factors
- Propose a **conditional modeling** of data
- Perform **classification** of unlabeled items

Warning: This LDA should not be confused with **Latent Dirichlet Allocation** (used in NLP)

Supervised Learning: Foundations

General Definition

Given data $X \subset \Omega \subseteq \mathbb{R}^D$ associated with categories (class labels) $Y = [[M]] \subset \mathbb{N}$.

Supervised learning consists of finding (learning) the parameters θ of a learner (function) ϕ_θ to minimize the loss $L_{X \times Y}(\theta)$ incurred when predicting class labels.

Prediction Function:

$$\phi_\theta : \Omega \to Y$$
$$x_i \mapsto \phi_\theta(x_i) = \tilde{y}_i$$

Examples of Learners

Logistic Regression:

ϕ_θ is a logistic regression with parameters θ

Neural Network:

ϕ_θ is a neural network with weights θ

Example Loss Function:

$$L_{X \times Y}(\theta) = \sum_X \|\phi_\theta(x_i) - y_i\|^2$$

Principle of LDA: Discriminant Direction

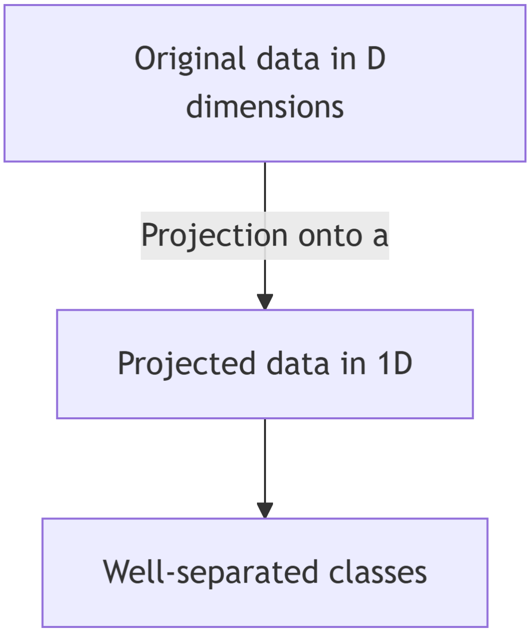
Problem Statement

Fundamental Question: Given multivariate data X associated with labels Y, we seek a model for this data.

Discriminant Direction with 2 Classes

We seek a **direction** $a \in \mathbb{R}^D$ such that the projection of the data onto this direction $\hat{X} = \mathrm{Proj}_a(X)$ shows **optimal properties for discrimination** between classes.

Geometric Visualization:



Key Principle: Find the direction a that best separates the classes after projection

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Between-Class Criterion

Principle of Maximum Separation

If the projections of the classes are **well separated**, then they can be easily discriminated.

Objective: Seek the direction a where between-class discrimination is **maximized**

Mathematical Formulation

Geometric Intuition:

The direction a should be **collinear** with the line $\mu_1 - \mu_2$ connecting the class centroids, because:

$$\|\hat{\mu}_1 - \hat{\mu}_2\|^2 = \left\| \frac{a^T \mu_1}{\|a\|^2} a - \frac{a^T \mu_2}{\|a\|^2} a \right\|^2 = \frac{1}{\|a\|^2} (a^T(\mu_1 - \mu_2))^2$$

This expression is **maximized** when $a = \lambda(\mu_1 - \mu_2)$

Important Point: The original data ($x_i \in X$) does not change. Only the **projected data** ($\hat{x}_i \in \hat{X}$) and subsequent calculations change as a evolves.

Property: The centroid of the projected data is the projection of the centroid of the original data

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Within-Class Criterion

Principle of Minimum Variance

We also seek to **minimize the variance** of each projected class individually.

Objective: Reduce dispersion within each class after projection

Within-Class Variance

For each class k, the normalized variance of the projection is:

$$\hat{\sigma}_k^2 = \frac{1}{N_k} \sum_{y_i = k} (\hat{x}_i - \hat{\mu}_k)^T (\hat{x}_i - \hat{\mu}_k)$$

Fisher's Criterion

Fisher's criterion maximizes the ratio:

$$\arg\max_a \frac{\|\hat{\mu}_1 - \hat{\mu}_2\|^2}{\hat{\sigma}_1^2 + \hat{\sigma}_2^2}$$

Interpretation:

- **Numerator:** Distance between projected means (between-class) → to **maximize**
- **Denominator:** Sum of within-class variances → to **minimize**

Generalization to M Classes

Generalized Between-Class Criterion

Definitions:

Let $B = [\mu_1 - \mu, \dots, \mu_M - \mu]$ be the matrix of centered class centroids, where:

- $\mu = \frac{1}{N}\sum_i^N x_i$ is the global mean
- $N = \sum_k N_k$ is the total number of points

Between-class covariance matrix:

$$\Sigma_b = \frac{1}{M} B B^T$$

Criterion to maximize:

$$\frac{1}{M}\sum_k (\hat{\mu}_k - \hat{\mu})^T(\hat{\mu}_k - \hat{\mu}) = \frac{1}{\|a\|^2} a^T \Sigma_b a$$

Generalized Within-Class Criterion

Definitions:

Let $A_k = [x_1 - \mu_k, \dots, x_{N_k} - \mu_k]$ with $y_i = k$, be the matrix of centered data for class k.

Within-class covariance matrix:

- For class k: $\frac{1}{N_k} A_k A_k^T$
- **Sum of within-class matrices:** $\Sigma_w = \sum_k \frac{1}{N_k} A_k A_k^T$

Criterion to minimize:

$$\sum_k \frac{1}{N_k} \sum_{y_i = k} (\hat{x}_i - \hat{\mu}_k)^T(\hat{x}_i - \hat{\mu}_k) = \frac{1}{a^T a} a^T \Sigma_w a$$

Development:

$$\sum_k \frac{1}{N_k} \sum_{y_i = k} \left[\frac{a^T(x_i - \mu_k)}{\|a\|^2} a\right]^T \left[\frac{a^T(x_i - \mu_k)}{\|a\|^2} a\right] = \frac{1}{a^T a} a^T \Sigma_w a$$

Combined Fisher's Criterion

Combining Both Criteria

Generalized Fisher's criterion combines the between-class and within-class criteria:

$$\arg\max_a L(a) = \arg\max_a \frac{a^T \Sigma_b a}{a^T \Sigma_w a}$$

Solution by Differentiation

To find the maximum, we compute the derivative:

$$\frac{\partial L(a)}{\partial a} = \frac{\Sigma_b a(a^T \Sigma_w a) - \Sigma_w a(a^T \Sigma_b a)}{(a^T \Sigma_w a)^2} = 0$$

Resulting Equation:

$$\Sigma_b a = L(a) \Sigma_w a$$

Solution: a is a solution of the **generalized eigenvalue problem**

The optimal direction a is the **first eigenvector** of $\Sigma_w^{-1} \Sigma_b$ (with eigenvalue $\lambda_1 = L(a)$)

Discriminant Subspaces

Multiple Discriminant Dimensions

Ordered Eigenvectors:

The eigenvectors corresponding to the **largest eigenvalues** λ_i are the most discriminative dimensions:

$$a_1, a_2, \dots, a_p \quad \text{with} \quad \lambda_1 > \lambda_2 > \dots > \lambda_p$$

Dimensional Limitation

Fundamental Property: M classes can be discriminated in a subspace of dimension **at most** $(M-1)$.

Consequence: There are only $M-1$ **non-zero** eigenvalues

Explanation: With M classes, we have M class centroids. Once globally centered, these centroids are constrained to lie in a space of dimension $M-1$.

Special Case: M = 2

For two classes, we can show that:

$$BB^T = (\mu_1 - \mu)(\mu_1 - \mu)^T + (\mu_2 - \mu)(\mu_2 - \mu)^T \propto (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

Implications:

- $BB^T a$ is a vector in the direction of $(\mu_1 - \mu_2)$
- The solution is $a = \lambda \Sigma_w^{-1}(\mu_1 - \mu_2)$

Practical Example: Character Recognition

Dataset Description

Partial MNIST Dataset:

- 7291 images of size 16×16 (8 bits)
- Digits from 0 to 9
- Representation: $\{x_i, y_i\}$ with $x_i \in \mathbb{R}^{256}$ and $y_i \in [[10]]$, $i \in [[7291]]$

Result of LDA Projection

LDA finds the **optimal subspace** for **linearly** separating the data according to their labels y_i.

Advantages of LDA Projection:

- Dimensionality reduction (from 256 to a few dimensions)
- Visualization in 2D or 3D possible
- Class separation maximized

Inference and Decision Making

Conditional Modeling

Classification Problem:

For a new data point j:

- x_j is **known**
- y_j is **unknown**

Question: To which class k does point j belong?

Probabilistic Approach

Declaration of a Categorical Random Variable:

We declare the class random variable C

Prediction by Bayes' Rule:

$$\mathbb{P}(C = k|x_j) = \frac{\mathbb{P}(x_j|C = k)\mathbb{P}(C = k)}{\mathbb{P}(x_j)}$$

Components:

- $\mathbb{P}(x_j|C = k)$: **likelihood**
- $\mathbb{P}(C = k)$: **prior** (a priori probability)
- $\mathbb{P}(x_j)$: **evidence** (normalization)

Gaussian Approximation

Class-Specific Gaussian Model

Distribution Assumption:

Each class is modeled by a normal distribution:

$$\mathbb{P}(x|C = k) \sim \mathcal{N}(\mu_k, \Sigma_k)$$

Prior Choice:

Uniform prior: $\mathbb{P}(C = k) = \frac{1}{M}$ (all classes equally probable)

Simplification:

The evidence $\mathbb{P}(x_j)$ is **ignored** as it is constant for all classes (normalization)

Maximum Likelihood

Approximation:

$$\mathbb{P}(x|C = k) \propto \exp\left(-(x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k)\right)$$

Decision Rule

Classification by Maximum A Posteriori:

$$\delta(x) = \arg\max_k \mathbb{P}(x|C = k)$$

Interpretation: We assign x to the class k that maximizes the likelihood (or posterior probability)

Optimality of LDA

Optimality Conditions

LDA is optimal when the M classes each follow a **Gaussian distribution**

Justification:

The discrimination criteria are based on covariance matrices Σ_w and Σ_b, which fully characterize Gaussian distributions.

Limitations

Linear Discriminant Analysis: does **not account** for **nonlinear** relationships between variables

Linear discrimination only

Alternative: For nonlinear relationships, use **nonlinear classification** methods:

- Kernel LDA
- Quadratic Discriminant Analysis (QDA)
- Support Vector Machines (SVM)
- Neural networks

Synthesis of Key Concepts

Essential Characteristics of LDA

Nature of the Method:

- LDA is a **supervised** technique
- Uses **class labels** to guide dimensionality reduction

Objective:

- Find the **optimal linear latent factors** that explain linear discrimination between classes

Fisher's Criterion:

- Combines **within-class** (within-scatter) and **between-class** (between-scatter) dispersion

- Maximizes the ratio $\frac{a^T \Sigma_b a}{a^T \Sigma_w a}$

Solution:

- Solved by finding the eigenvalues of the matrix $\Sigma_w^{-1}\Sigma_b$
- The eigenvectors form the discriminant axes

Applications:

- **Visualization** of data considering labels
- **Supervised dimensionality reduction**
- **Classification** via conditional modeling (each class follows a normal distribution)

Comparison: LDA vs PCA

Fundamental Differences

PCA (Unsupervised):

- Maximizes **global variance**
- Ignores class labels
- Finds directions of maximum variance
- Based on Σ (total covariance)

LDA (Supervised):

- Maximizes **separation between classes**
- Uses class labels
- Finds discriminant directions
- Based on $\Sigma_w^{-1}\Sigma_b$ (ratio of covariances)

Complementarity

In Practice: PCA and LDA can be combined

1. PCA for initial dimensionality reduction (eliminate noise)
2. LDA to maximize discrimination

Typical Exam Questions

Conceptual Questions

Supervised Learning Setup:

- Explain the context of supervised learning
- Difference between supervised and unsupervised learning

Linear Nature of LDA:

- Why is LDA a linear method?
- What are its limitations?

Discriminant Direction:

- What characterizes a discriminant direction?
- What does the vector a represent?

Technical Questions

Fisher's Criteria:

- What is Fisher's criterion?
- How does it combine between-class and within-class criteria?

Between-Class Criterion:

- Principle of the between-class criterion
- How to compute the between-class covariance matrix Σ_b?

Within-Class Criterion:

- Principle of the within-class criterion
- How to compute the within-class covariance matrix Σ_w?

Mathematical Proofs

Derivation of the Maximum:

- Justify and explain the derivation of the maximum for a
- Show that a satisfies $\Sigma_b a = \lambda \Sigma_w a$

Multidimensional Situation:

- Describe the multidimensional situation ($D > 2$)
- Explain why there are at most $M - 1$ discriminant dimensions

Inference:

- How to perform inference with LDA?
- Provide an example of conditional modeling using LDA

Mathematical Concepts to Master

To understand LDA well, you need to master:

Linear Algebra:

- Eigenvectors and eigenvalues
- Generalized eigenvalue problem
- Covariance matrices
- Orthogonal projection

Statistics:

- Mean and variance
- Covariance and correlation
- Multivariate normal distribution
- Bayes' rule

Optimization:

- Differentiation of matrix functions
- Constrained optimization
- Rayleigh quotient method